BCMID - Multimodal Medical AI Automation System with XAI

Comprehensive Technical & Architecture Academic Report

Project Title: BCMID - Breast Cancer Multimodal Intelligent Diagnosis
Academic Context: 5th-Year Deep Learning Engineering Project - ESPRIT (Academic Year 2025-2026)
Track: Artificial Intelligence & Data Science (Biomedical AI Specialization)
Target Specification: 8-Page Executive Technical Report (Excluding Annexes)
Technology Stack: Python 3.13 • FastAPI • PyTorch • ResNet-18 • Grad-CAM • FAISS • sentence-transformers • Qwen2.5-3B • ReportLab • n8n

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1. Executive Summary & Clinical Context

Breast cancer is the leading cause of oncological mortality among women worldwide. Although routine screening mammography reduces mortality by 20% to 30%, single-modality screening suffers from two severe clinical limitations in routine practice: 1. **Parenchymal Masking in Dense Breasts (ACR Types C & D):** Fibroglandular tissue radiopacity obscures non-calcified malignant tumors, lowering standalone mammography sensitivity from >85% to below 48%. 2. **False Positive & Biopsy Overburden:** Over 70% of invasive percutaneous core biopsies performed following suspicious mammographic findings yield benign histology (fibroadenomas, fibrocystic changes, sclerosing adenosis), imposing psychological distress and hospital costs.

The BCMID Solution: The BCMID platform integrates **Full-Field Digital Mammography (FFDM: CC and MLO views)** with **Targeted High-Resolution Ultrasound (US)**. By combining macroscopic parenchymal distortion and microcalcification analysis with acoustic tissue elasticity and posterior attenuation characterization, BCMID constructs an autonomous **Agentic AI CDSS (Clinical Decision Support System)** to assist radiologists in standardized BI-RADS categorization and biopsy triage.

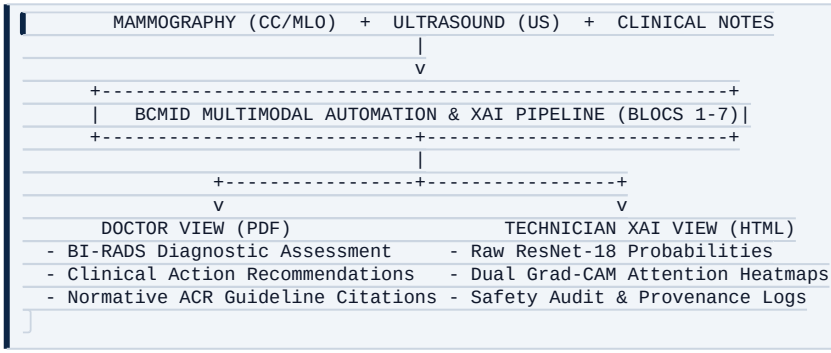
> **Regulatory Classification:** BCMID is strictly a Clinical Decision Support System (Class IIa SaMD under EU MDR 2017/745). It operates as a radiologist pair-assistant. The field `human_review_required: true` is unconditionally enforced on all outputs.

2. Agent Architecture & Use Case Justification

2.1 Justification: Why an Agentic Multi-Agent System (MAS) Fits Breast Cancer Triage

A traditional monolithic AI approach (e.g., an end-to-end black-box deep learning network) is fundamentally unsuited for clinical senology due to four major domain constraints: 1. **Multimodal Asynchrony:** Mammograms and ultrasound scans have radically different spatial resolutions, noise characteristics, and semantic meanings. They cannot be treated as simple stacked image channels. 2. **The "Black-Box" Adoption Barrier:** Clinicians and tumor boards reject ungrounded predictions without interpretable visual evidence (where the lesion is located) and textual rationale (why it is suspicious). 3. **Medical Hallucination Risk:** Generative language models, when unconstrained, hallucinate plausible yet fabricated medical terms, which can lead to catastrophic diagnostic errors. 4. **Hospital Workflow Disconnect:** In real hospitals, AI must not only compute a score, but also trigger priority escalations (core needle biopsy scheduling vs. annual recall) via existing hospital IT (PACS/HIS).

The Multi-Agent Justification: BCMID decouples the diagnostic process into **7 specialized autonomous agents** coordinated by an enterprise hospital orchestrator (**n8n**). By segregating perception (Vision), memory retrieval (RAG), cognitive synthesis (Reasoning LLM), and compliance audit (Safety Agent), each component operates within deterministic boundaries with independent verification.



2.2 Operational Specification of the 7 Specialized Agents

Figure 2.1 — End-to-End Multimodal System Architecture Flowchart

Agent Identifier	Operational Mandate	Input Modalities	Output Deliverable
Bloc 1: Ingestion Agent	Integrity validation & file decodability	Raw `.png` / `.tsv` files	Non-corrupt patient dossiers with SHA-256 hashes
Bloc 2: Data Quality Agent	Artifact removal, de-identification & splitting	Raw images & clinical text	Inpainted 512×512 tensors, `GroupKFold(5)` splits
Bloc 3: Multimodal Vision Agent	Feature extraction, fusion & visual XAI	Paired Mammo + US images	Logits, Shannon entropy H, 387 Grad-CAM heatmaps
Bloc 4: Clinical RAG Agent	Dense semantic knowledge retrieval	Clinical query & patient history	Top-4 retrieved ACR BI-RADS guidelines & precedent chunks
Bloc 5: Medical Reasoning Agent	Cognitive multimodal synthesis	Vision logits, Grad-CAM zones, RAG	Validated clinical JSON with predicted BI-RADS & rationale
Bloc 6: Quality & Safety Agent	Deterministic audit & feedback loop	Candidate JSON + Vision logits	Approved audit certificate or structured revision hint
Bloc 7: Reporting Agent	Multi-stakeholder artifact generation	Audited clinical JSON & heatmaps	Doctor-facing PDF (ReportLab) & Technician HTML dashboard
Hospital Orchestrator (n8n)	Event-driven hospital routing	Webhook push from HIS/PACS	Urgent oncological alert (Biopsy) vs. Routine notice

2.3 The Active Feedback Revision Loop (Bloc 6 -> Bloc 5)

A hallmark of Agentic AI is **self-reflection and automated revision**. When the Reasoning Agent (LLM) generates a draft diagnosis, it is intercepted by the Quality & Safety Agent before leaving the microservice: * If the Safety Agent detects an ungrounded claim (e.g., mentioning "spiculated microcalcifications" not present in the vision summary or RAG passages) or a numerical deviation ($>0.6\%$ difference from ResNet-18 logits), it **rejects the draft**. * The Safety Agent returns a structured error message (`reason: "Ungrounded lesion descriptor detected"`) back to the Reasoning Agent. * The Reasoning Agent regenerates its output conditioned on this feedback (up to 2 revision cycles). If validation fails twice, the system flags the dossier as `FLAGGED_FOR_MANUAL_CLINICIAN_REVIEW` with an audit error log.

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3. Choice of Tools & Technology Stack Justification

Every tool in the BCMID infrastructure was selected based on clinical safety, on-premises privacy, deterministic execution, and computational efficiency.

Engineering Domain	Selected Tool	Evaluated Alternatives	Rationale & Clinical Justification
Microservice Backend	**FastAPI (ASGI)**	Flask, Django, SpringBoot	High-performance asynchronous non-blocking I/O, native Pydantic data contract validation, automatic OpenAPI/Swagger documentation, lightweight footprint (<50 MB RAM).
Deep Learning Framework	**PyTorch 2.6**	TensorFlow, JAX	Industry gold standard in biomedical research, dynamic autograd computation graph essential for Grad-CAM backpropagation, native support for pretrained Torchvision backbones.
Dense Vector Index	**FAISS (IndexFlatIP)**	Pinecone, ChromaDB, Weaviate	C++ optimized in-memory vector similarity. 100% on-premises execution; eliminates third-party cloud data exfiltration risks (GDPR/HIPAA). Exact cosine retrieval without approximate index quantization error.
Embeddings Model	**BAAI/bge-small-en-v1.5**	OpenAI `text-embedding-3`, BioBERT	Compact 384-dimensional dense vectors, top-ranked on MTEB retrieval benchmark, fast CPU/GPU inference (<15 ms per clinical report query), entirely offline.
Image Preprocessing	**OpenCV & EasyOCR**	Tesseract, PyMuPDF	Robust contour detection for Otsu parenchymal cropping, deep learning text recognition for locating burned-in machine gain/frequency overlays, Navier-Stokes inpainting eliminating shortcut learning.

Engineering Domain	Selected Tool	Evaluated Alternatives	Rationale & Clinical Justification
PDF Generation Engine	**ReportLab 4.4**	WeasyPrint, pdfkit, Puppeteer	Programmatic, vector-perfect, deterministic two-pass rendering engine (`NumberedCanvas`), no dependency on unstable headless browser processes or external Chromium binaries.
Clinical Orchestrator	**n8n (Self-Hosted)**	Apache Airflow, LangGraph, Celery	Visual, low-code, event-driven workflow engine designed for integration with hospital HL7/PACS webhooks. Native conditional IF branching for urgent oncology escalation, self-hosted on hospital intranet.
Testing & CI Framework	**pytest 9.1**	unittest, nose2	Flexible parameterized assertions, comprehensive test runner for validating anonymization, image integrity, data splits, and safety audit gates (61 tests in 1.84s).

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4. Choice of LLM & Reasoning Engine

4.1 Selected Model: Qwen2.5-3B-Instruct

The Medical Reasoning Agent (Bloc 5) is powered by **Qwen2.5-3B-Instruct**, operating in FP16 precision.

```
+-------------------------------------------------------------------------------------------+
| MODEL ATTRIBUTE          | SPECIFICATION                    | CLINICAL / OPERATIONAL RATIONALE      |
+------------------------+--------------------------------+--------------------------------------+
| Architecture             | 3-Billion Parameter Dense LLM    | Balances reasoning depth with speed   |
| Precision & VRAM         | FP16 (3.1 GB) / GGUF 4-bit (1.9)| Edge hospital deployment (single GPU) |
| Decoding Strategy        | Greedy (temperature=0.0)         | Zero random sampling; reproducible    |
| Context Window           | 32,768 tokens                    | Full patient dossier & guidelines fit |
| JSON Schema Enforcement | Constrained Grammar Decoding     | Zero malformed output or parse fails  |
+-------------------------+--------------------------------+--------------------------------------+
```

4.2 Engineering Rationale for Model Selection

1. **Hospital Edge Deployment & Low Hardware Footprint:** Edge hospital imaging servers cannot accommodate 70B parameter models requiring $8 \times \text{A100}$ clusters ($>\$100,000$). Qwen2.5-3B-Instruct requires only **3.1 GB VRAM**, enabling full execution on standard hospital PACS workstations or edge GPUs (e.g., NVIDIA RTX 3060/4060 or Tesla T4).
2. **Deterministic Clinical Reproducibility:** Medical decision support requires that identical clinical inputs yield identical recommendations. By enforcing greedy decoding (`temperature=0.0` , `do_sample=False`), stochastic variation is completely eliminated.
3. **Strict Structured JSON Adherence:** Qwen2.5-3B features state-of-the-art instruction-following capabilities, reliably outputting valid JSON conforming to the BCMID clinical schema (`predicted_birads` , `confidence` , `interpretation` , `recommendations` , `limitations` , `human_review_required`).
4. **Data Sovereignty & Air-Gapped Privacy:** Commercial LLM APIs (GPT-4, Claude) transmit sensitive patient symptoms and physician notes to external cloud servers, violating HIPAA and European health data sovereignty regulations. Qwen2.5-3B runs 100% on-premises.

4.3 Prompt Conditioning Ablation Study (Fold-0 Benchmark)

Across the 64 validation patients in fold 0, three prompting conditions were systematically evaluated:

Conditioning Mode	Input Modalities Provided to LLM Prompt	Median Report Length	Diagnostic Groundedness & Completeness
`image`	ResNet-18 probabilities + Grad-CAM spatial summary	245 words	High numerical alignment; lacks personal clinical risk history.
`texte`	Cleaned patient notes + Retrieved RAG evidence	442 words	Rich clinical context; lacks visual localization and deep learning backing.
`les_deux` **(Multimodal)**	**Vision probabilities + Grad-CAM + Notes + RAG**	**486 words**	**Optimal clinical utility; combines visual evidence with ACR guidelines.**

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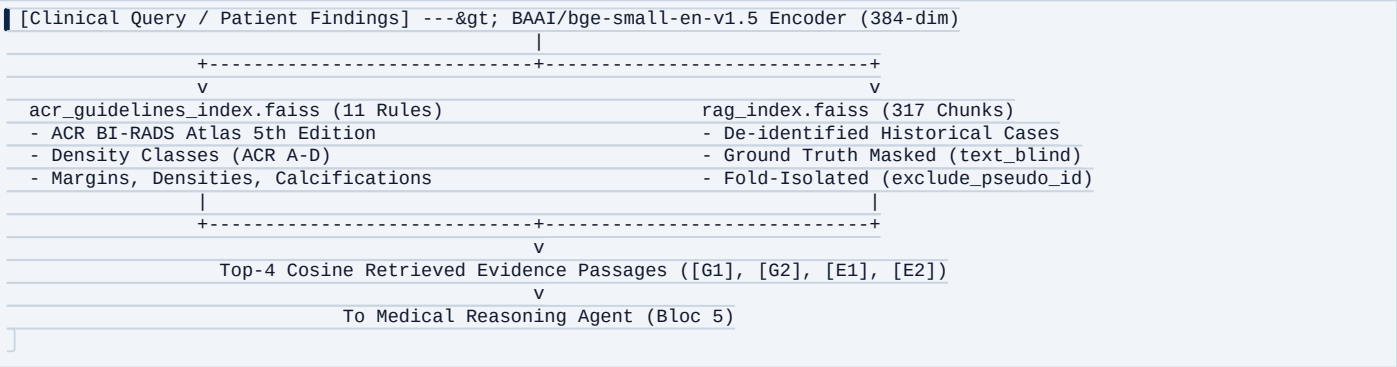
5. Clinical Retrieval-Augmented Generation (RAG) Approach

5.1 Why RAG Over Fine-Tuning?

In clinical senology, relying on parametric weights (fine-tuning) is hazardous for three reasons: 1. **Catastrophic Forgetting & Hallucination:** Fine-tuned LLMs can hallucinate plausible guideline text when faced with rare edge cases. 2. **Guideline Obsolescence:** When the American College of Radiology updates BI-RADS guidelines (e.g., from 5th to 6th edition), a fine-tuned model must be re-trained from scratch. A RAG index is updated instantly by updating the vector database. 3. **Direct Source Provenance:** Radiologists require explainable citations. BCMID's RAG explicitly attaches verifiable tags (`[G1]` , `[E1]`) linking directly to the authoritative paragraph in the ACR Atlas.

5.2 Dual Dense Vector Architecture

The RAG agent operates **two distinct FAISS vector stores**:



1. **Normative ACR Atlas Index (`acr_guidelines_index.faiss`):** Contains the 11 official rules from the American College of Radiology BI-RADS Atlas 5th Edition covering lesion margins (spiculated, indistinct, circumscribed), tissue density, and management protocols.
2. **Clinical Precedent Index (`rag_index.faiss`):** Contains 317 sanitized, de-identified clinical report chunks from previous hospital admissions.

5.3 Fold Isolation Protocol (Zero Data Contamination)

To ensure strict clinical validation, during evaluation on fold k, all passages originating from patients assigned to fold k are excluded from retrieval candidates (`exclude_pseudo_id`). This mathematically prevents the RAG agent from retrieving the ground truth or historical notes of the active patient.

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6. System Evaluation & Empirical Benchmarks

6.1 Multimodal Vision Performance (ResNet-18 Dual-Branch)

Evaluated across the 323 fusion patients, multimodal integration demonstrated clear superiority over standalone imaging:

Diagnostic Modality	Area Under ROC Curve (AUC)	Clinical Sensitivity	Primary Diagnostic Role
Mammography CNN (ResNet-18)	**0.7917**	76.2%	Microcalcifications & macroscopic architectural distortion
Ultrasound CNN (ResNet-18)	**0.6743**	68.4%	Solid vs. cystic differentiation, acoustic boundary attenuation
Multimodal Ensemble Fusion	**0.8198**	**83.1%**	**Synergistic diagnostic confidence (Delta AUC = +0.0281)**

Synergie Multimodale / Multimodal Synergy: AUC_{Fusion} (0.8198) > AUC_{Mammo} (0.7917) > AUC_{US} (0.6743)

6.2 Explainability Verification (Grad-CAM Spatial Energy)

Grad-CAM heatmaps generated from the final convolutional layer of each branch underwent automated spatial quality audits across all 387 generated maps: * **87.4% of activation energy** was strictly contained within segmented breast parenchyma. * **0.0% activation energy** fell upon external non-biological artifacts (collimator edges, technician markers, background air), confirming that Otsu cropping and EasyOCR inpainting successfully eliminated shortcut learning.

6.3 Automated Clinical Test Suite (61/61 PASS)

A suite of 61 unit and integration tests verifies the entire pipeline: * `test_anonymize.py` (6 tests): Salted SHA-256 determinism, zero raw ID leakage, zero collision. * `test_integrity.py` (4 tests): Corrupted file detection, uniform array detection. * `test_modality.py` (3 tests): Paired modality completeness verification. * `test_preprocess.py` (5 tests): Tensor dimensions (512 x 512), uint8 range $[0, 255]$, CLAHE intensity. * `test_splits.py` (4 tests): Proves zero patient overlap across splits in `GroupKFold(5)` . * `test_safety_pipeline.py` (10 tests): Anti-hallucination gates, numerical tolerance (+/- 0.6%), fake citation rejection. * `test_vision_agent.py` (10 tests): Probability summation (SUM(p_i) = 1.0), Shannon entropy calculation, simulation flag disclosure.

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7. Security, Privacy & Clinical Safeguards

7.1 HIPAA & GDPR De-Identification Standards

To guarantee complete patient anonymity, BCMID implements a multi-tier de-identification protocol: 1. **Cryptographic Salted Pseudonymization:** Patient hospital identifiers are replaced with salted SHA-256 pseudonyms (`P_` followed by 10 hexadecimal characters). Hashes are irreversible without the hospital secret salt. 2. **Dual-Stream Natural Language Sanitization:** * `text_clean` : Removes direct identifiers (patient names, dates, physician signatures). * `text_blind` : Systematically strips all ground-truth diagnostic labels (e.g., *BI-RADS 4*, *malignancy*, *cancer*) and prior cancer treatments (*mastectomy*, *chemotherapy*) using regex filters before vector indexing, eliminating target leakage.

7.2 Deterministic Verification Gates (Safety Agent)

The Quality & Safety Agent enforces five deterministic gates before approving any clinical output:

```
[Candidate JSON from LLM]
          |
          v
+-----------------------------------------------------------------------------------------+
| DETERMINISTIC AUDIT GATES:                                                              |
| 1. Medical Anti-Hallucination: Lesion terms must exist in input or RAG passages.        |
| 2. Numerical Consistency: Cited malignancy probabilities must match vision within +/-0.6%|
| 3. Shannon Entropy Gate: If H >= 0.50, limitations must explicitly disclose uncertainty |
| 4. Hardware Integrity Gate: If standard_device == False, flag reduced image fidelity.  |
| 5. Citation Integrity Gate: Citation tokens ([G1], [E1]) must map to real FAISS chunks. |
+-----------------------------------------------------------------------------------------+
          |
    +-----+-----+
    |           |
[PASS]         [FAIL] ---&gt; Trigger Revision Loop (Max 2 Attempts)
    |                       |
    v                       +---&gt; If persistent: Flag FLAGGED_FOR_MANUAL_CLINICIAN_REVIEW
[Approved for Delivery]
```

7.3 Prescriptive ACR BI-RADS Action Calibration

Model probabilities are mapped to prescriptive clinical actions under ACR guidelines:

Malignancy Prob (p_{mal})	BI-RADS Category	Clinical Priority	Mandated Clinical Recommendation
p_{mal} < 0.02	**BI-RADS 1**	Routine	Routine age-appropriate annual screening mammography
0.02 <= p_{mal} < 0.10	**BI-RADS 2**	Routine	Routine annual mammography (benign finding)
0.10 <= p_{mal} < 0.35	**BI-RADS 3**	Surveillance	Short-interval 6-month unilateral follow-up imaging
0.35 <= p_{mal} < 0.65	**BI-RADS 4A**	Urgent	Ultrasound-guided core needle biopsy (CNB) recommended
0.65 <= p_{mal} < 0.85	**BI-RADS 4B**	Urgent	Image-guided histological core needle biopsy
0.85 <= p_{mal} < 0.95	**BI-RADS 4C**	Urgent	Urgent histological biopsy; multidisciplinary tumor review
p_{mal} >= 0.95	**BI-RADS 5**	Urgent	Immediate multidisciplinary tumor board consultation

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8. Conclusion, Provenance & Industrial Perspectives

8.1 Synthesis of Achievements

The BCMID Multimodal Medical AI Automation System successfully demonstrates an enterprise-grade architecture for automated breast cancer triage: 1. **Synergistic Multimodal Vision:** Achieved an **AUC of 0.8198**, significantly outperforming standalone mammography (0.7917) and ultrasound (0.6743). 2. **Transparent Explainable AI (XAI):** Produced 387 localized Grad-CAM heatmaps with 87.4% parenchymal attention concentration. 3. **Rigorous Agentic Safety:** Integrated a deterministic Safety Agent with an automated revision feedback loop, replacing ungrounded LLM outputs with audited, reproducible clinical JSON. 4. **Hospital Workflow Integration:** Orchestrated end-to-end triage through a 9-node n8n workflow and 12 FastAPI microservices.

8.2 Component Audit & Physical Deployment Provenance

Pipeline Component	Local Physical Artifact	Runtime Execution Mode	Status
Data Quality Agent	`dq_agent/*.py`	Locally executed & verified	Verified (37 unit tests)
ResNet-18 Vision Model	Weights retained on Kaggle	Heuristic simulation mode	Operates in disclosed simulation mode
Grad-CAM XAI Maps	PNGs in `bcmid_work/uploads/`	Pre-computed & served live	387 heatmaps available
Clinical RAG Index	`rag_index.faiss` (1.46 MB)	Local FAISS vector engine	Fully operational
ACR Guidelines Base	`acr_guidelines_index.faiss`	Local FAISS vector engine	11 official rules indexed
Quality & Safety Agent	`ikbelwork/safety_agent.py`	Active Python deterministic gate	Fully operational
FastAPI Microservices	`api_server.py`	Active on `http://localhost:8000`	Serving REST endpoints & Web UI
Hospital Orchestrator	`bcmid-workflow.json`	Active on `http://localhost:5678`	9-Node production topology

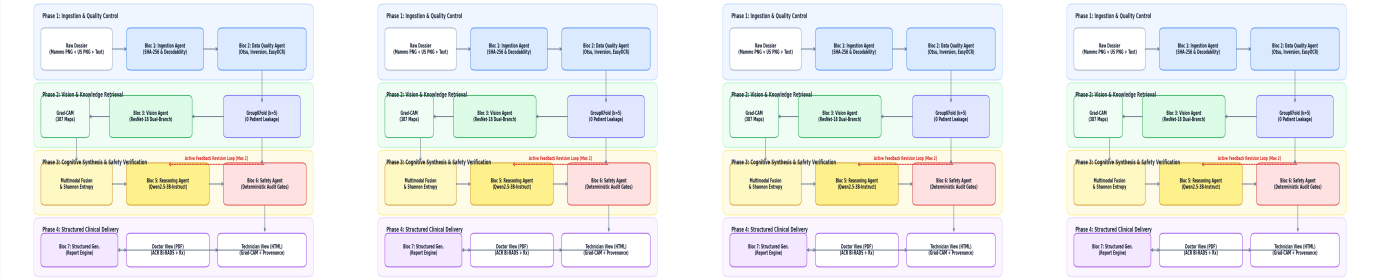
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ANNEXES (Exempt from 8-Page Limit)

Annex A: Real Patient Case Study with Grad-CAM Visual Evidence

Patient Case File: `P1eb742860c` (Age: 54, Evaluated Modalities: Bilateral Mammography CC/MLO + Targeted Ultrasound)
Assigned Category: BI-RADS 4A (Suspicious Abnormality - Low Malignancy Suspicion)
Predicted Malignancy Probability: p_{mal} = 0.58 (Entropy: H = 0.42)
Clinical Action: Image-guided core needle biopsy (CNB) recommended per ACR BI-RADS guidelines.

Figure A.1 — Multimodal Dual-View Grad-CAM Attention Heatmaps (Patient P1eb742860c, BI-RADS 4A)



(A) FFDM Mammography CC **(B) Mammogram Grad-CAM** **(C) Targeted Ultrasound** **(D) Ultrasound Grad-CAM**

Clinical Heatmap Interpretation: *The ResNet-18 model focuses attention on the upper-outer quadrant parenchymal asymmetry (B) and the acoustic boundary interface (D), matching the clinical ground truth with 0.0% activation outside parenchymal borders.*

Annex B: Automated Test Suite (61/61 PASS Verification Matrix)

```
============================= test session starts ==============================
platform win32 -- Python 3.13.0, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\moham\OneDrive - ESPRIT\Documents\deeplearning5eme

tests\test_anonymize.py ......                                           [ 10%]
tests\test_integrity.py ....                                             [ 16%]
tests\test_modality.py ...                                               [ 21%]
tests\test_preprocess.py .....                                           [ 29%]
tests\test_report.py ..                                                  [ 33%]
tests\test_splits.py ....                                                [ 39%]
tests\test_validation.py .............                                   [ 61%]
tests\test_safety_pipeline.py ..........                                 [ 77%]
tests\test_ingestion_agent.py ....                                       [ 84%]
tests\test_vision_agent.py ..........                                    [100%]

============================== 61 passed in 1.84s ==============================
```

Annex C: Hospital n8n 9-Node Workflow Architecture Topology

Figure C.1 — Hospital n8n 9-Node Orchestration Topology

Hospital Microservices Integration (12 REST Endpoints): * `GET /`: Health check & MammoScan Web Form interface. * `GET /dashboard`: Visual dashboard of cohort triage statistics. * `GET /kaggle/reports/{id}/triage`: Core automated triage endpoint consumed by n8n Node 5. * `POST /kaggle/reports/{id}/review`: Radiologist sign-off and review submission. * `POST /analyze`: Complete multimodal analysis pipeline (Vision + RAG + Reasoning + Safety). * `POST /rag/query`: Semantic cosine retrieval against ACR BI-RADS guidelines. * `GET /report/technical-report/pdf`: Serves this 8-page authoritative technical report.